

Faithful two-pole closure for the viscous drop modes

Reid's arbitrary-Oh eigenvalues mapped onto the drop-mode ODE

SpectralKM derivations

1 What is being matched, and why

Each droplet surface mode in this model is carried by a second-order ODE (`eq:lambda-1`; implemented in `src/affine.jl:drop_affine`)

$$\ddot{\beta}_l + 2\lambda_l \dot{\beta}_l + \omega_l^2 \beta_l = F_l b_l, \quad (1)$$

whose characteristic polynomial is $s^2 + 2\lambda_l s + \omega_l^2$. The published choice takes Lamb's (1881) small-viscosity asymptotics,

$$\lambda_l^{\text{Lamb}} = \text{Oh} (l-1)(2l+1), \quad (\omega_l^{\text{Lamb}})^2 = \omega_{l,0}^2 = l(l-1)(l+2), \quad (2)$$

with $\omega_{l,0}$ the inviscid Rayleigh–Lamb capillary frequency. Reid (1960) instead gives the *exact* linear viscous eigenvalues at arbitrary Oh, as roots σ of a transcendental characteristic equation (see `src/reid.jl`); with σ a decay rate, perturbations behave as $e^{-\sigma t}$, so the ODE roots are $s = -\sigma$.

This note fixes the remaining question: *given a pair of Reid eigenvalues, what are the faithful (λ_l, ω_l^2) ?* The algebra is verified symbolically in `derivations/cas_reid_two_pole.py` (all assertions pass).

2 The mapping is Vieta, and needs no case split

For a monic quadratic with roots s_1, s_2 ,

$$(s - s_1)(s - s_2) = s^2 + 2\lambda_l s + \omega_l^2 \iff \boxed{\lambda_l = -\frac{1}{2}(s_1 + s_2), \quad \omega_l^2 = s_1 s_2} \quad (3)$$

exactly, for *any* pair. Two consequences.

Underdamped ($s = -\gamma \pm i\omega$).

$$\lambda_l = \gamma, \quad \omega_l^2 = \gamma^2 + \omega^2, \quad (4)$$

not ω^2 . The distinction matters because (1) oscillates at $\sqrt{\omega_l^2 - \lambda_l^2}$, and only (4) returns ω . Lamb (2) carries the same $O(\lambda^2/\omega^2)$ slip, but there it sits inside its own asymptotic error; here it would be gratuitous.

Overdamped ($s = -\gamma_1, -\gamma_2$, **real**, $\gamma_1 < \gamma_2$).

$$\lambda_l = \frac{1}{2}(\gamma_1 + \gamma_2), \quad \omega_l^2 = \gamma_1 \gamma_2, \quad \lambda_l^2 - \omega_l^2 = \left(\frac{\gamma_1 - \gamma_2}{2}\right)^2 \geq 0, \quad (5)$$

so the ODE is genuinely overdamped, degenerating only when the rates coincide. As $\gamma_2 \rightarrow \gamma_1$ this reduces continuously to the critically damped case, hence (3) is a strict generalisation.

3 The superseded choice, and why it was badly wrong

An earlier revision of `drop_viscous_coeffs` set, for a real root,

$$\lambda_l = \gamma_{\text{tracked}}, \quad \omega_l^2 = \lambda_l^2, \quad (6)$$

a double root at the tracked rate. This discards γ_2 and invents a second rate. Two compounding errors made it worse than a mild approximation:

1. **Wrong branch.** Continuation in Oh from the complex pair does *not* land on the slowest real root once the pair has merged. Measured at $l = 2$ by scanning the real axis (the residual is real for real $s > 0$, since $q = \sqrt{s/\text{Oh}}$ is then real and so is j_l):

Oh	real roots on ($10^{-4}, 200$)	tracked	slowest
0.8	2.040, 3.670, 23.94, 63.04, 118.3, 189.4	3.670	2.040
1.0	1.280, 5.846, 29.93, 78.80, 147.9	5.846	1.280
3.0	0.357, 20.97, 89.86	20.97	0.357

Continuation consistently returns the *second* root. The slowest root decreases with Oh (as a very viscous drop must relax ever more slowly, rate $\propto 1/\text{Oh}$), whereas the tracked one grows — the diagnostic that exposed this.

2. **Wrong stiffness.** Applying (3) to the two slowest roots gives $\omega_l^2 = 7.487, 7.485, 7.480$ at Oh = 0.8, 1.0, 3.0, against $\omega_{2,0}^2 = l(l-1)(l+2) = 8$. The faithful mapping therefore **preserves the capillary stiffness**, essentially independent of Oh, as it must: viscosity damps, it does not change what restores. The superseded choice (6) instead gave $\omega_l^2 = \lambda_l^2 = 20.97^2 \approx 440$ at Oh = 3, a stiffness too large by a factor 59.

That $\omega_l^2 \approx \omega_{l,0}^2$ is not an input to the construction; it emerges, and so serves as a cheap correctness check on the root pair.

4 Where the overdamped branch is actually reached

The capillary pair merges onto the real axis (mode becomes aperiodic) at an Oh that falls with l : measured between 0.5 and 0.8 for $l = 2, 4$, and between 0.3 and 0.5 for $l = 8, 16$. So the overdamped case is not hypothetical for Oh $\gtrsim 0.5$, and high modes enter it first.

5 Algorithm

For each l , at the requested Oh:

1. Track the least-damped eigenvalue by continuation in Oh from a value small enough that Lamb is genuinely asymptotic (`reid_root_tracked`).
2. If $|\text{Im } \sigma| > 0$: the pair is $\sigma, \bar{\sigma}$; apply (4).
3. If $\text{Im } \sigma = 0$: the pair has merged. Scan the positive real axis for roots (real residual, bisection on sign changes, rejecting pole crossings at zeros of j_l), take the *two slowest*, apply (5). Do *not* rely on the continued root being the slowest.
4. Sanity-check ω_l^2 against $\omega_{l,0}^2$; a large discrepancy indicates a bad root pair rather than physics.

6 Scope

Two limitations are inherited and unchanged by this note. First, (1) is a two-pole model: Reid's spectrum also contains an infinite diffusive ladder (internal Stokes relaxations, rates $\approx \text{Oh } q_j^2$ with q_j the zeros of j_l), so the *forced* response carries memory that no second-order ODE reproduces. Matching two poles exactly is right for free decay and approximate under forcing; the n -pole generalisation is deferred. Second, the residues are not used here — only the poles — so this note says nothing about the amplitude with which pressure forcing excites each mode.